MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

1/ TYPE 1G3GTA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, half-wave, high vacuum

Outline	---	See figure 1
Envelope	---	T9
Base	---	B6-8, B8-6, or B8-58
Cap	---	C1-1, C1-34, or C1-49
Cathode	---	Coated filament
Base connections:		See note 1

Pin No.	---	1	2	3	4	5	6	7	8	Cap
Element	---	int con	f	int con	nc	int con	nc	f, int sd	int con	a

ABSOLUTE-MAXIMUM RATINGS:

RF service:

Parameter:	Ef	epx	ib	Io	F	Zp	X-radiation	Alt
Unit:	Vdc	kv	ma	mAdc	kHz	---	mr/hr	ft
Maximum:	1.45	33	30	1.1	100	Note 2	0.5 (see note 4)	10,000 (see note 3)
Minimum:	1.05	---	---	---	1.5	---	---	---

Flyback service: See note 5

Parameter:	Ef	epx	ib	Io	F	Rℓ	Cℓ	X-radiation	Alt
Unit:	Vac	kv	ma	mAdc	kHz	meg	pF	mr/hr	ft
Maximum:	1.45	26	50	0.5	100	---	---	0.5 (see note 4)	10,000 (see note 3)
Minimum:	1.05	---	---	---	1.5	---	---	---	---
TEST CONDITIONS: (See note 5)	0.95	26	35	0.34	---	60	2,000	---	---

GENERAL:

Qualification - Required

1/ Replaces tube type 1B3GT

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Qualification inspection								
1216	Base material insulating quality	-	Zone 5 (min)	---	---	---	---	---	---
	Quality conformance inspection, part 1								
1301	Filament current	-	Ef = 1.25 V	0.65	II	If	180	220	mA
---	Output current	6		0.65	II	Io	0.275	---	mAdc
1231	Emission current	-	Eb = 100 Vdc; Ef = 1.10 Vac; Rℓ = Cℓ = 0	0.65	II	Is	5.0	15.0	mAdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
---	High-voltage rectifier breakdown	7		2.5	II	---	---	---	---
1261	Electrode voltage (anode)	-	Eb (Vdc)/Ib = 7.0 mAdc; Ef = 1.25 V; Rℓ = Cℓ = 0	---	---	Etd	70	130	Vdc
1331	Direct-interelectrode capacitance	-		6.5	Code E	Cp	1.0	2.0	pF
2202	X-radiation	-	Ef = 0; Eb = -26 kVdc	4.0	Code B	X-rad	---	0.5	mr/hr
1111	Base pin solder depth	-		---	---	---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life	-	Group A	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):								
1231	Emission current	-		---	---	Is	3.0	---	mAdc
---	Output current	-		---	---	Io	0.215	---	mAdc
2202	X-radiation	-		---	---	X-rad	---	0.5	mr/hr

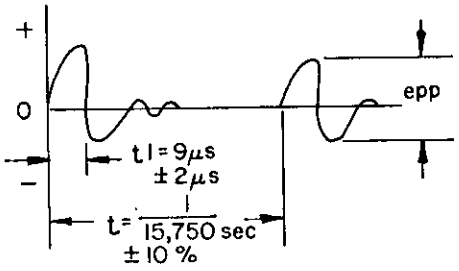
NOTES:

1. Connecting pins 1, 3, 5, and 8 to pin 7 externally is permissible to reduce corona discharge; otherwise pins 1, 3, 5, and 8 may not be used. Pins 4 and 6 may be omitted. Socket terminals 4 and 6 may be used as tie points at or near filament potential.
2. The supply source impedance when used at maximum voltage and current ratings shall be 105,000 ohms, or greater.
3. If altitude rating is exceeded, reduction of instantaneous voltage may be required.
4. Based on accumulated sample test data, taken initially and during life test, tubes do not exceed this maximum rating limit of 0.5 mr/hr at any time throughout their useful life when operated within the absolute maximum ratings, including filament voltage, specified herein.

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NOTES: -Continued

5. The anode supply voltage waveform is to be as shown below.



6. This test to be performed at the conclusion of the holding period.

7. The tube shall be operated under the specified conditions for 6 seconds before tapping. Breakdown during the first 3 seconds of the "no tap" interval is permissible; however, breakdown during the last 3 seconds shall be cause for rejection.

   (a) Breakdown during "no tap" interval:

      (1) Continuous or intermittent arcing anywhere within the tube.

      (2) The tube shall then be tapped in sequence of 25 taps each (approximate time per sequence = 5 seconds). A tube which survives a full sequence without a breakdown shall pass the test. When a breakdown occurs during a sequence, the sequence shall be discontinued and another sequence begun. The tube shall be rejected if a breakdown occurs during two successive sequences.

   (b) Breakdown during tapping interval;

      (1) Any fluctuation of Io greater than 10 microamperes.

      (2) Continuous or intermittent arcing anywhere within the tube.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

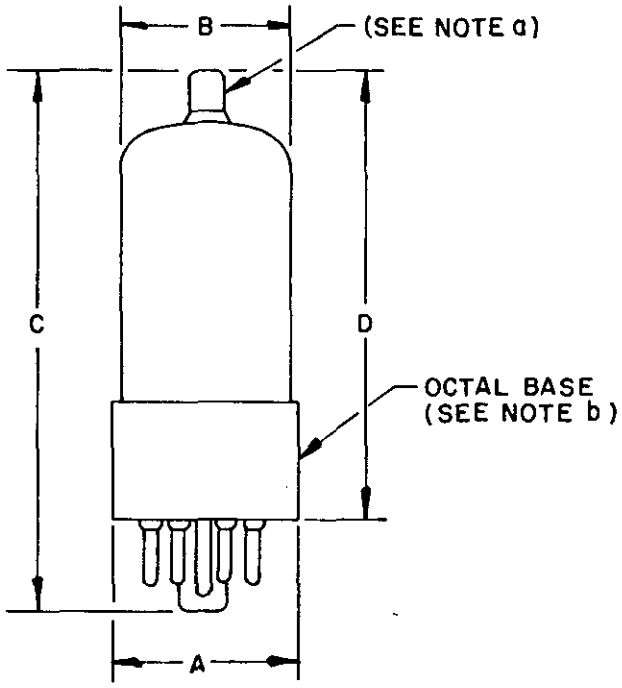
User activities:
Army - WC
Navy - AS, MC, CG

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3000)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	1.235 (31.37)	1.275 (32.39)
B	1.062 (27.00)	1.188 (30.10)
C		3.563 (90.50)
D	2.625 (66.68)	3.000 (76.20)

NOTES:
a. Connectors should not exert more than 7 pounds radial compression at any point around the circumference of the cap.
b. Pins 4 and 6 may be omitted.

FIGURE 1. Outline drawing of electron tube type 1G3GTA.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
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